invert the metaphor: we ride the river of time as it relentlessly rushes forward, sweeping us from one now to the next, as the past recedes with the passing scenery and the future forever awaits us downstream. (Our experiences have also taught us that time can inspire some of the mushiest metaphors.) Second, *time seems to have an arrow.* The flow of time seems to go one way and only one way, in the sense that things happen in one and only one temporal sequence. If someone hands you a box containing a short film of a glass of milk being spilled, but the film has been cut up into its individual frames, by examining the pile of images you can reassemble the frames in the right order without any help or instruction from the filmmaker. Time seems to have an intrinsic direction, pointing from what we call the past toward what we call the future, and things appear to change—milk spills, eggs break, candles burn, people age—in universal alignment with this direction.

These easily sensed features of time generate some of its most tantalizing puzzles. Does time really flow? If it does, what actually is flowing? And how fast does this time-stuff flow? Does time really have an arrow? Space, for example, does not appear to have an inherent arrow—to an astronaut in the dark recesses of the cosmos, left and right, back and forth, and up and down, would all be on equal footing—so where would an arrow of time come from? If there is an arrow of time, is it absolute? Or are there things that can evolve in a direction opposite to the way time's arrow seems to point?

Let's build up to our current understanding by first thinking about these questions in the context of classical physics. So, for the remainder of this and the next chapter (in which we'll discuss the flow of time and the arrow of time, respectively) we will ignore quantum probability and quantum uncertainty. A good deal of what we'll learn, nevertheless, translates directly to the quantum domain, and in Chapter 7 we will take up the quantum perspective.

Does Time Flow?

From the perspective of sentient beings, the answer seems obvious. As I type these words, I clearly *feel* time flowing. With every keystroke, each now gives way to the next. As you read these words, you no doubt feel